

Management Information Systems

Class Notes: Classes 24-32

Part II • Sections MIS-3 (cont.), MIS-4, MIS-5 • IS & Org, Telecommunications, DSS & EIS

CO2 — MIS in Business Strategy & Decision Making

Bloom's: Understand → Analyze → Apply

Semester Final: 72 Marks

CONTENTS

- 24 Roles of IT & IS in Organizational Decision Making
- 25 Systems Approach to Problem Solving
- 26 Telecommunications & the Information Superhighway
- 27 Signals, Communication Channels & Media
- 28 Network Topology, Software & Communication Types
- 29 Telecom Network Types, Carriers & Business Value
- 30 Information & Levels of Management
- 31 Information & Decision Making
- 32 Decision Support Systems (DSS)

Roles of IT & IS in Organizational Decision Making

CO2

Bloom's: Analyze

Teaching: Lecture + Visual + Exercise

1. Decision Making Levels

Organizational decisions occur at three distinct levels, each with different information needs, time horizons, and IS support requirements:

Level	Who Makes Decisions	Time Horizon	Information Needed	IS Type
Strategic (Top)	CEO, Board, C-Suite	Long-term (3–10 years)	Aggregate, external, future-oriented, unstructured	EIS/ESS, DSS
Tactical (Middle)	Department heads, regional managers	Medium-term (months–1 year)	Summarized, semi-structured, departmental performance	MIS, DSS
Operational (Lower)	Supervisors, team leaders, frontline staff	Short-term (daily–weekly)	Detailed, real-time, structured, transaction-level	TPS, Operational MIS

2. Decision Types by Structure

Structured Decisions

Routine, repetitive decisions with clear rules and procedures. The same inputs always produce the same correct output. No managerial judgment required — can be fully automated by IS.

Examples: approving a purchase order under \$10,000, calculating employee overtime pay, reordering inventory when stock falls below reorder point.

Semi-Structured Decisions

Part of the problem is structured (analyzable with data and models), but part requires human judgment. IS supports but does not replace the decision maker.

Examples: setting a product's selling price, evaluating a supplier's bid, determining the optimal marketing budget allocation across channels.

Unstructured Decisions

Novel, complex decisions with no defined procedure. High uncertainty, multiple conflicting objectives. IS provides information support; human judgment dominates.

Examples: deciding whether to enter a new international market, choosing a merger target, responding to an unexpected competitive threat, crisis management.

EXAM TIP

Map decision type to management level: Operational = Structured; Tactical = Semi-structured; Strategic = Unstructured. And map IS type to decision type: TPS/MIS handles structured; DSS handles semi-structured; EIS supports unstructured. Questions often ask you to classify a given scenario correctly across all three dimensions simultaneously.

3. Simon's Model of Decision Making

HERBERT A. SIMON — NOBEL LAUREATE IN ECONOMICS (1978)

Herbert Simon's model describes decision making as a four-phase process. IS supports each phase differently.

○ Phase 1 — Intelligence

"What is the problem?" Scanning the environment to identify problems and opportunities requiring a decision. IS role: MIS exception reports, EIS dashboards, TPS anomaly detection, competitive intelligence IS. Example: the MIS exception report flags that sales in Region B dropped 18% — this triggers the intelligence phase for the sales manager.

○ Phase 2 — Design

"What are the options?" Developing, analyzing, and evaluating alternative courses of action. IS role: DSS what-if models, simulation tools, financial planning models. Example: the sales manager uses a DSS to model three alternatives — price reduction, increased advertising, or new product launch — and projects the revenue impact of each.

○ Phase 3 — Choice

"Which option is best?" Selecting the best alternative from those developed in the Design phase. IS role: optimization models, decision matrices, sensitivity analysis in DSS. The manager selects "increased advertising" based on the DSS analysis showing highest projected ROI.

○ Phase 4 — Implementation

"How do we execute and monitor?" Putting the chosen solution into effect and monitoring results. IS role: TPS executes the operational changes; MIS monitors KPIs; feedback loops detect if results match expectations and trigger a new intelligence phase if not.

4. How TPS → MIS → DSS → EIS Data Flow Works

The IS hierarchy is not just a classification — data flows upward through the layers, transforming at each stage:

Flow	What Is Passed	Transformation	Example
TPS → MIS	Raw transaction records	Aggregation, summarization, period-end totalling	10,000 daily sales transactions → "Weekly sales by product category by region" report
MIS → DSS	Summarized reports + historical trends	Model input, what-if parameters, trend data for forecasting	3 years of weekly sales reports → DSS demand forecasting model for next quarter's production plan
DSS → EIS	Analytical outputs, scenario results	Further aggregation, KPI extraction, graphical presentation	DSS scenario analysis → EIS dashboard showing "Best/Base/Worst case profit scenarios for Board review"

5. Group Decision Making

GDSS CONCEPTS

DEFINITION

A **Group Decision Support System (GDSS)** is an IS that supports group decision making by providing structured tools for idea generation, evaluation, voting, and consensus-building — enabling geographically dispersed participants to collaborate effectively.

- ▶ **Electronic brainstorming:** All participants simultaneously submit ideas through the IS without interrupting each other — eliminates the "loud voice dominates" problem of face-to-face meetings.
- ▶ **Anonymous input:** Participants can submit ideas, evaluations, and votes without their identity being revealed — reduces social pressure, hierarchy effects, and groupthink.
- ▶ **Voting and ranking:** GDSS aggregates votes and displays ranked results instantly — no manual tallying, no ambiguity.
- ▶ **Parallel processing:** All participants contribute simultaneously rather than sequentially — dramatically reduces meeting time.

NOMINAL GROUP TECHNIQUE (NGT)

A structured group decision method where members independently generate ideas (written, not spoken), then share one at a time in a round-robin format, followed by group discussion and secret voting. IS supports NGT by collecting and displaying written submissions anonymously, aggregating votes, and displaying ranked results.

DELPHI METHOD

A multi-round expert consultation technique for reaching consensus on uncertain questions (e.g., technological forecasting, long-range planning). Each round, experts submit anonymous written opinions; the IS compiles a summary and statistical summary of responses; experts then revise their views for the next round based on the group's aggregate response. Rounds continue until convergence. IS enables remote, asynchronous Delphi rounds — no need to assemble experts physically.

6. IS for Competitive Intelligence

- ▶ **Environmental scanning IS:** Continuously monitors news feeds, social media, regulatory databases, and competitor websites to detect early signals of market change. Example: Grameenphone's IS monitors regulatory announcements from BTRC in real-time.
- ▶ **Monitoring IS:** Tracks specific competitors' pricing, product launches, and job postings (a hiring surge in a competitor's R&D department signals an upcoming product launch).
- ▶ **Forecasting IS:** Uses historical data and models to project future market conditions, demand levels, and competitor moves.

7. IS in Supply Chain and Inter-Organizational Decisions

- ▶ **Supply chain coordination IS:** Shares demand forecasts, inventory levels, and production schedules with suppliers — enabling them to plan production without waiting for formal purchase orders. Reduces lead times and inventory buffers throughout the supply chain.
- ▶ **Inter-organizational IS (IOS):** IS that spans organizational boundaries, connecting a firm with its suppliers, customers, distributors, or logistics partners. Enabled by EDI, APIs, or shared cloud platforms.
- ▶ **Example — Bangladesh RMG sector:** Major Bangladeshi garment factories (Beximco, DBL Group) use IS connected to buyers' (H&M, Zara) procurement systems — automatically receiving purchase orders, updating production schedules, and confirming shipment details without phone or email communication.

1. Systems Thinking — Holistic vs. Reductionist View

Dimension	Reductionist View	Systems (Holistic) View
Approach	Break the problem into parts; solve each part independently	Study the whole system and the interactions between parts
Assumption	The whole equals the sum of its parts	The whole is more than the sum of its parts (emergent properties)
Risk	Solving one part may create problems in another (sub-optimization)	Changes are evaluated for their effect on the entire system
Example	IT department optimizes the server for maximum throughput — but the network becomes the bottleneck	The entire IS infrastructure is reviewed together — server, network, and storage balanced as a system

WHY THIS MATTERS FOR MIS

When implementing an MIS, a reductionist approach might optimize the database for speed without considering the impact on the reporting module, or redesign the sales process without considering how it integrates with the inventory IS. Systems thinking requires evaluating every change in the context of the whole organization's information flow.

2. The System Approach — Six Steps

○ Step 1 — Define the Problem

Precisely identify what is wrong — not the symptoms, but the root cause. Example: "Our monthly financial reports are taking 15 days to produce" is a symptom. The problem (root cause) may be: "Data from three separate systems must be manually reconciled because there is no integrated IS." Defining the wrong problem leads to solving the wrong thing.

○ Step 2 — Analyze the System

Understand the current system — its components, boundaries, inputs, outputs, processes, and constraints. Map data flows, identify bottlenecks, measure performance gaps. Use techniques like Data Flow Diagrams (DFDs), entity-relationship models, and process mapping.

○ Step 3 — Identify Alternatives

Generate multiple potential solutions. For an MIS problem: (a) buy a packaged ERP, (b) develop custom IS in-house, (c) upgrade existing system with integration middleware, (d) outsource to a cloud-based SaaS MIS. Each alternative should be distinct and feasible.

○ **Step 4 — Select the Best Solution**

Evaluate alternatives against criteria (cost, time, risk, functionality, scalability) and select the most appropriate solution. May use multi-criteria decision analysis (MCDA), NPV comparison (see below), or risk-weighted scoring.

○ **Step 5 — Implement the Solution**

Execute the chosen solution — for IS, this means following the SDLC (System Development Life Cycle): design, build, test, deploy, train users, cut over from old system.

○ **Step 6 — Evaluate the Results**

Compare actual outcomes against expected outcomes. Did the IS solve the root cause problem? Are KPIs improving? Is user adoption satisfactory? Evaluation feeds back into the cycle — if the problem persists, return to Step 1 with new information.

3. Problem vs. Symptom — Root Cause Analysis

THE 5-WHY TECHNIQUE

The 5-Why technique is a root cause analysis method where you repeatedly ask "Why?" until the underlying cause is identified. Developed by Sakichi Toyoda and used extensively in Toyota's production system.

5-WHY EXAMPLE — MIS CONTEXT

Symptom: The management report was wrong this month.

Why 1: Because the sales figures were incorrect. → *Why?*

Why 2: Because the sales team entered some transactions twice. → *Why?*

Why 3: Because the IS allowed duplicate entries — no uniqueness constraint on transaction IDs. → *Why?*

Why 4: Because the IS was designed 8 years ago without proper database constraints. → *Why?*

Why 5: Because there was no data quality review in the original IS design process — no database administrator was involved.

Root Cause: Inadequate IS design governance — no DBA involvement and no data integrity controls.

Solution: add uniqueness constraints + duplicate-detection validation + appoint a DBA to review IS design.

FISHBONE (ISHIKAWA) DIAGRAM

The fishbone diagram is a cause-and-effect visualization tool. The "head" of the fish is the problem; the "bones" represent categories of potential causes (for IS: People, Process, Technology, Data, Environment, Management). Each bone is explored for

contributing factors. Useful for group problem-solving sessions and for systematically covering all potential root causes before jumping to solutions.

4. Feasibility Study

Before committing to an IS solution, a feasibility study evaluates whether the proposed system is viable across four dimensions:

Technical Feasibility

Does the technology to build this IS exist? Does the organization have (or can it acquire) the necessary hardware, software, and technical expertise?

Example: "Can we build a real-time IoT-integrated inventory IS with our current infrastructure?" — if the factory has no internet connectivity in the warehouse, it is not technically feasible without major infrastructure investment.

Economic Feasibility

Do the expected benefits exceed the costs? This is the most rigorous feasibility dimension — requires formal cost-benefit analysis using NPV (see below).

Benefits: cost savings + revenue gains + productivity improvement. **Costs:** hardware, software, development, training, maintenance, disruption.

Operational Feasibility

Will the IS actually be used and work within the organization's operational environment? Are there enough trained users? Does it fit existing workflows? Is there management commitment?

Example: A sophisticated DSS deployed in an organization where managers have no data analysis skills and distrust computer-generated recommendations — technically feasible but operationally infeasible.

Schedule Feasibility

Can the IS be developed and deployed within the required time window? Are deadlines driven by business events (regulatory deadline, competitor launch) that cannot be moved?

Example: A tax compliance IS must be operational before the January tax filing deadline — if the development team estimates 18 months and the deadline is 10 months away, it is not schedule feasible.

5. Economic Feasibility — NPV Calculation

NET PRESENT VALUE (NPV)

NPV calculates the present value of an IS project's future cash flows, discounted at the organization's cost of capital. A positive NPV means the project creates value; a negative NPV means it destroys value.

$$NPV = \sum [B_t \div (1 + r)^t] - C_0$$

where: B_t = net benefit in year t / r = discount rate / t = year / C_0 = initial cost

NPV WORKED EXAMPLE — IS PROJECT EVALUATION

A garment export company considers implementing a Supply Chain MIS. Initial investment: ₹80,00,000 (Co). Expected annual net benefits: ₹25,00,000/year for 5 years. Discount rate (cost of capital): 10%.

Year-by-year discounting:

Year 1: $25,00,000 \div (1.10)^1 = 25,00,000 \div 1.10 = \mathbf{22,72,727}$

Year 2: $25,00,000 \div (1.10)^2 = 25,00,000 \div 1.21 = \mathbf{20,66,116}$

Year 3: $25,00,000 \div (1.10)^3 = 25,00,000 \div 1.331 = \mathbf{18,78,287}$

Year 4: $25,00,000 \div (1.10)^4 = 25,00,000 \div 1.464 = \mathbf{17,07,534}$

Year 5: $25,00,000 \div (1.10)^5 = 25,00,000 \div 1.611 = \mathbf{15,52,303}$

Sum of PV of benefits = ₹94,76,967

NPV = 94,76,967 – 80,00,000 = ₹14,76,967 (Positive → Project is financially viable)

Interpretation: Even accounting for the time value of money, this IS project generates ₹14.77 lakh of value above its cost over 5 years. Management should approve it on economic grounds.

6. Soft Systems Methodology (SSM)

PETER CHECKLAND'S SSM

Soft Systems Methodology (SSM) is an approach for tackling complex, ill-defined (unstructured) organizational problems where there is no clear agreement on what the problem is or what a solution looks like. Developed by Peter Checkland at Lancaster University. Unlike the hard systems approach (which assumes a clear problem and a single best solution), SSM treats problem definition itself as part of the process.

- ▶ **Suitable for:** MIS implementation challenges, organizational change decisions, problems with strong political and cultural dimensions — where different stakeholders have genuinely different views of what "the problem" is.
- ▶ **Key concept — CATWOE:** A structured technique for defining a system. Customers (who benefits?), Actors (who performs the activities?), Transformation (what does the system do?), Weltanschauung (what worldview makes this system meaningful?), Owner (who controls the system?), Environmental constraints (what limits the system?).
- ▶ **Process:** Explore the real world → create "rich pictures" (messy diagrams of stakeholders, concerns, and relationships) → define relevant systems → build conceptual models → compare models to reality → identify desirable and feasible changes → take action.

7. IS Development Lifecycle (SDLC)

○ Phase 1 — Planning

Define project scope, objectives, and feasibility. Secure management approval and budget. Form the project team. Produce the Project Initiation Document (PID).

○ Phase 2 — Analysis

Gather and document requirements — what must the IS do? Techniques: user interviews, observation, document analysis, workshops. Produce a Requirements Specification document.

○ Phase 3 — Design

Specify how the IS will fulfil the requirements — database design (ER diagrams), system architecture (client-server, cloud), user interface wireframes, data flow diagrams, security design.

○ Phase 4 — Implementation

Build the IS — coding, database creation, hardware procurement, network configuration, integration testing. Includes user acceptance testing (UAT) where actual users validate the system against requirements.

○ Phase 5 — Maintenance

Ongoing operation — bug fixes, performance tuning, security patches, enhancements as business needs evolve. The longest and most expensive phase over the IS's lifetime (typically 60–80% of total IS cost).

Telecommunications & the Information Superhighway

CO2

Bloom's: Understand

Teaching: Lecture + Diagram + Visual

1. Telecommunications — Definition

DEFINITION

Telecommunications is the electronic transmission of signals for communications over distances, including data, voice, and video, through various media such as copper wire, fiber optic cable, microwave, satellite, and radio waves. It is the physical and logical infrastructure over which IS components communicate.

2. The Information Superhighway

DEFINITION

The **Information Superhighway** is the concept of a global digital communication infrastructure — high-bandwidth, ubiquitous, interconnected networks — that enables the rapid transmission of vast amounts of information between individuals, organizations, and governments worldwide. The Internet is the primary realization of this concept.

The metaphor of a "superhighway" captures key properties: high capacity (bandwidth), multiple lanes (simultaneous users), on-ramps and off-ramps (access points), and connectivity across vast distances. Just as physical highways enabled the movement of goods, the information superhighway enables the movement of data — the raw material of MIS.

3. Components of a Telecommunications Network

Terminals

Input/output devices at the user end of the network — the points where users send and receive information.

Examples: computers, smartphones, tablets, POS terminals, ATMs, IoT sensors, printers, smart TVs.

Communication Channels

The physical or wireless medium over which data travels — the "road" of the telecommunication network.

Examples: twisted pair copper wire, coaxial cable, fiber optic cable, radio waves, microwave links, satellite links.

Communication Processors

Communication Software

Hardware devices that control, route, and manage the flow of data through the network.

Examples: routers (direct packets between networks), switches (connect devices within a LAN), modems (convert digital ↔ analog), multiplexers, bridges.

Programs that manage the network, implement protocols, and provide services to applications.

Examples: network operating system (Windows Server, Linux), protocol stacks (TCP/IP), network management software, VPN software, firewall software.

4. Telecommunications Protocols

DEFINITION

A **telecommunications protocol** is a set of rules and standards that define how data is formatted, transmitted, received, and acknowledged between communicating devices. Without protocols, devices from different manufacturers could not communicate.

Protocol	Full Name	Layer	Purpose
TCP/IP	Transmission Control Protocol / Internet Protocol	Transport + Network	The foundational protocol suite of the Internet. IP handles addressing and routing; TCP ensures reliable, ordered delivery of data packets.
HTTP/HTTPS	HyperText Transfer Protocol (Secure)	Application	Transfers web pages and data between web servers and browsers. HTTPS adds TLS encryption for security — essential for MIS web interfaces and e-commerce.
FTP	File Transfer Protocol	Application	Transfers files between computers over a TCP/IP network. Used for bulk data transfer, software distribution, and IS data exchange between organizations.
SMTP	Simple Mail Transfer Protocol	Application	Sends email messages between mail servers. MIS exception reports and alerts are often delivered via SMTP-based email systems.
DNS	Domain Name System	Application	Translates human-readable domain names (mis.company.com) to IP addresses (192.168.1.10). Without DNS, users would need to memorize IP addresses.

5. OSI Reference Model — 7 Layers

The **Open Systems Interconnection (OSI)** model, developed by ISO, provides a conceptual framework for understanding how network communications work, divided into 7 layers. Each layer has specific responsibilities and communicates only with the layers immediately above and below it.

7	Application	User-facing protocols and services. HTTP, FTP, SMTP, DNS, SNMP. This is where MIS web applications, email, and file transfer operate. End-user software interacts at this layer.
6	Presentation	Data translation, encryption, and compression. Converts data from application format to network format (e.g., ASCII to EBCDIC). Handles encryption (TLS operates here conceptually) and data compression. Ensures that data sent by one application can be read by another.
5	Session	Session management. Establishes, maintains, and terminates communication sessions between applications. Manages dialogue control (half-duplex vs. full-duplex) and synchronization. Example: ensures that a database transaction session stays open until all operations are complete.
4	Transport	End-to-end reliable data delivery. TCP: guaranteed, ordered delivery with error checking and retransmission. UDP: fast, unreliable delivery (used for streaming, DNS queries). Handles segmentation of large messages into packets and reassembly at destination.
3	Network	Logical addressing and routing. IP protocol operates here — assigns IP addresses and determines the best path (route) for packets to travel from source to destination across multiple networks. Routers operate at this layer.
2	Data Link	Node-to-node data transfer and error detection. Packages raw bits into frames; adds MAC (Media Access Control) addresses for local delivery. Handles error detection (CRC). Ethernet, Wi-Fi (802.11) operate at this layer. Switches operate at Layer 2.
1	Physical	Physical transmission of raw bits. Defines electrical/optical signals, voltage levels, cable specifications, connector types, data rates. Copper wire, fiber optic, radio waves operate at this layer. Hubs and repeaters operate at Layer 1.

EXAM TIP — OSI MNEMONIC

Remember OSI layers top-to-bottom (7→1): **"All People Seem To Need Data Processing"** (Application, Presentation, Session, Transport, Network, Data Link, Physical). Or bottom-to-top (1→7): **"Please Do Not Throw Sausage Pizza Away."** Exam questions typically ask you to: (1) list all 7 layers in order, (2) identify what protocol/device operates at a given layer, (3) describe the function of a specific layer.

6. TCP/IP Model — 4-Layer Mapping

TCP/IP Layer	Equivalent OSI Layers	Key Protocols
Application	OSI 5 (Session) + 6 (Presentation) + 7 (Application)	HTTP, HTTPS, FTP, SMTP, DNS, SSH, SNMP
Transport	OSI 4 (Transport)	TCP (reliable), UDP (fast/unreliable)
Internet	OSI 3 (Network)	IP (IPv4, IPv6), ICMP, ARP
Network Access	OSI 1 (Physical) + 2 (Data Link)	Ethernet, Wi-Fi (802.11), ARP, MAC addressing

7. IP Addressing — IPv4 and IPv6

- ▶ **IPv4 format:** XXX.XXX.XXX.XXX — four octets (8 bits each), written in decimal. Example: 192.168.1.105. Total address space: $2^{32} = 4,294,967,296$ (~4.3 billion) unique addresses.
- ▶ **IPv4 exhaustion:** The Internet Assigned Numbers Authority (IANA) allocated the last IPv4 address blocks in 2011. The explosion of internet-connected devices (smartphones, IoT sensors) exhausted the supply. NAT (Network Address Translation) has extended IPv4's life by allowing many private devices to share one public IP address.
- ▶ **IPv6:** 128-bit addresses written in hexadecimal (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334). Address space: $2^{128} \approx 3.4 \times 10^{38}$ — virtually unlimited. Designed to permanently solve the address exhaustion problem and support IoT scale (billions of connected devices).

IPv4 address space: **232** = 4,294,967,296 addresses (~4.3 billion) IPv6 address space: **2128** $\approx 3.4 \times 10^{38}$ addresses (≈ 340 undecillion)

1. Analog vs. Digital Signals

Property	Analog Signal	Digital Signal
Nature	Continuous waveform — varies smoothly over time	Discrete values — only 0 or 1 (binary)
Representation	Sine wave with amplitude, frequency, and phase	Square wave — high voltage = 1, low voltage = 0
Noise sensitivity	High — noise degrades signal quality; hard to distinguish from the original wave	Low — noise rarely changes a 0 to a 1 if margins are adequate; can be regenerated perfectly
Transmission distance	Degrades over distance (attenuation); requires amplifiers	Regenerated at repeaters — retransmitted at full strength, no quality degradation
Examples	Traditional telephone (PSTN), AM/FM radio, vinyl records	Computer data, fiber optic transmission, digital telephone (VoIP), Ethernet
Bandwidth efficiency	Lower — continuous waveform uses spectrum continuously	Higher — compression and error correction highly effective with digital data

FROM ANALOG TO DIGITAL — SAMPLING AND QUANTIZATION

- ▶ **Sampling:** Measuring the amplitude of an analog signal at regular time intervals. The sampling rate must be at least twice the highest frequency in the signal (Nyquist sampling theorem) to accurately reconstruct the signal.
- ▶ **Quantization:** Rounding each sample to the nearest value in a finite set of levels. An 8-bit quantization scheme has $2^8 = 256$ levels; 16-bit has 65,536 levels (CD audio quality).
- ▶ **Encoding:** Converting the quantized values into binary (0s and 1s) for digital transmission.

2. Modulation

When digital data must be transmitted over an analog medium (like traditional telephone lines), it must be converted to an analog signal — this is **modulation**. The reverse (analog to digital) is **demodulation**. A **modem** (MODulator-DEModulator) performs both.

AM — Amplitude Modulation

FM — Frequency Modulation

The amplitude (height) of the carrier wave is varied to represent data. High amplitude = 1, low amplitude = 0.

Used in: AM radio broadcasting, some older data modems. Susceptible to noise (which affects amplitude).

The frequency of the carrier wave is varied. Higher frequency = 1, lower frequency = 0.

Used in: FM radio, audio broadcasting, older telephone modems (FSK — Frequency Shift Keying). More noise-resistant than AM.

PM — Phase Modulation

The phase of the carrier wave is shifted to represent data. Phase shift of 180° = change in bit value.

Used in: modern high-speed modems, Wi-Fi (QPSK, QAM), satellite communications. Enables very high data rates when combined with amplitude modulation (QAM).

3. Channel Bandwidth and Maximum Data Rate

BANDWIDTH

Bandwidth is the range of frequencies a communication channel can carry, measured in Hertz (Hz). Greater bandwidth = more data can be transmitted per second. In everyday usage, bandwidth is often used loosely to mean data throughput (bits per second), but technically it refers to the frequency range.

NYQUIST'S THEOREM — MAXIMUM DATA RATE (NOISELESS CHANNEL)

Max Data Rate = $B \times \log_2(M)$ bits/second

where: B = channel bandwidth in Hz | M = number of discrete signal levels

NYQUIST THEOREM — WORKED EXAMPLE

A noiseless telephone channel has a bandwidth of 3,000 Hz, and the modulation scheme uses 4 signal levels ($M = 4$).

Max Data Rate = $3,000 \times \log_2(4) = 3,000 \times 2 = 6,000$ bits/second (6 kbps)

If we increase to 8 signal levels ($M = 8$): Max Data Rate = $3,000 \times \log_2(8) = 3,000 \times 3 = 9,000$ bps

Insight: Doubling the number of signal levels increases the maximum data rate, but requires better signal quality and more sophisticated hardware to distinguish between more levels reliably.

SHANNON'S THEOREM — MAXIMUM DATA RATE (NOISY CHANNEL)

Max Data Rate = $B \times \log_2(1 + S/N)$ bits/second

where: B = bandwidth in Hz | S/N = Signal-to-Noise Ratio (linear, not dB)

SHANNON'S THEOREM — WORKED EXAMPLE

A telephone channel: bandwidth = 3,000 Hz; Signal-to-Noise Ratio = 1,000 (30 dB).

Max Data Rate = $3,000 \times \log_2(1 + 1,000) = 3,000 \times \log_2(1,001) \approx 3,000 \times 9.97 \approx 29,900 \text{ bps} \approx 30 \text{ kbps}$

This is why traditional dial-up modems plateaued at ~56 kbps — they were bounded by Shannon's theorem given telephone channel bandwidth and noise levels.

4. Transmission Media

Medium	Type	Speed	Max Distance	Cost	Use Case
Twisted Pair (Cat 5e)	Guided (wired)	Up to 1 Gbps	100 m per segment	Very low	LAN connections within offices and buildings; telephone wiring
Twisted Pair (Cat 6/6A)	Guided (wired)	Up to 10 Gbps	55–100 m	Low	High-speed LAN, data center interconnects
Coaxial Cable	Guided (wired)	Up to 10 Gbps	500 m–1 km	Medium	Cable TV (broadband Internet), earlier Ethernet (10Base2/5)
Fiber Optic	Guided (optical)	Up to 100+ Tbps	Tens of km	High (installation)	Backbone networks, long-distance telecom, data center interconnects, ISP infrastructure
Radio Waves	Unguided (wireless)	Varies (Wi-Fi: up to 9.6 Gbps with Wi-Fi 6)	Varies (tens of meters to km)	Low (infrastructure)	Wi-Fi (LAN), Bluetooth (PAN), cellular networks
Microwave	Unguided (wireless)	Up to several Gbps	50+ km per link (line-of-sight)	Medium	Long-distance telecom backbone, telecom towers, rural connectivity
Satellite	Unguided (wireless)	Up to 100+ Mbps	Global	High	Remote area connectivity, GPS, maritime/aviation, Starlink broadband

5. Communication Processors

- ▶ **Modem:** Converts digital signals to analog (modulation) for transmission over analog telephone lines, and converts analog back to digital (demodulation) at the receiving end. Essential for DSL broadband connections.
- ▶ **Multiplexer (MUX):** Combines multiple signals from several sources onto a single high-capacity channel for efficient transmission. A demultiplexer separates them at the destination. Enables bandwidth sharing — instead of 10 separate cables for 10 connections, one high-capacity cable carries all 10 multiplexed signals.
- ▶ **Router:** Operates at OSI Layer 3 (Network). Reads IP addresses in packets and forwards them toward their destination across different networks. Routers make the Internet work — they determine the best path from source to destination across millions of possible routes.
- ▶ **Switch:** Operates at OSI Layer 2 (Data Link). Connects devices within a LAN and forwards frames only to the specific port where the destination MAC address is connected — far more efficient than a hub (which broadcasts to all ports).
- ▶ **Hub:** A simple Layer 1 device — repeats incoming signals to all connected ports. All devices share the same bandwidth. Largely obsolete, replaced by switches.
- ▶ **Bridge:** Connects two network segments at Layer 2, filtering traffic so only necessary frames cross the bridge. Reduces unnecessary traffic and separates collision domains.

6. Channel Characteristics

Characteristic	Definition	Impact on MIS
Bandwidth	Frequency range the channel can carry (Hz) / colloquially, data capacity (bps)	Higher bandwidth allows more MIS data to be transmitted simultaneously — essential for real-time dashboards and video conferencing
Latency	Time delay for a signal to travel from sender to receiver (milliseconds)	High latency makes real-time IS (ATM, stock trading, remote surgery IS) unusable. Satellite links have 500–600ms latency — problematic for interactive MIS
Throughput	Actual data transfer rate achieved in practice (bps) — always $\leq$ theoretical bandwidth	Throughput determines how fast MIS can exchange data with remote branches or cloud servers
Noise	Unwanted electrical or electromagnetic interference that corrupts the signal	High noise causes data errors — MIS must implement error detection (CRC) and correction to maintain data integrity

Characteristic	Definition	Impact on MIS
Attenuation	Progressive weakening of signal strength over distance	Limits the maximum cable length without repeaters; fiber optic has very low attenuation, enabling long-distance IS connectivity

Network Topology, Software & Communication Types

CO2

Bloom's: Understand

Teaching: Lecture + Demo + Diagram

1. Network Topologies

A **network topology** is the physical or logical arrangement of nodes and connections in a network. The choice of topology affects cost, performance, fault tolerance, and scalability.

Topology	Structure	Advantages	Disadvantages	Use Case
Bus	All devices connected to a single shared cable (backbone). Data broadcast to all; only intended recipient accepts it.	Simple, cheap, easy to extend	Single point of failure (break anywhere = network down); performance degrades with more devices; difficult to troubleshoot	Legacy Ethernet (10Base2); small temporary networks
Star	All devices connect to a central hub or switch. All communication goes through the centre.	Easy to add/remove devices; fault in one cable doesn't affect others; easy to troubleshoot	Entire network fails if the central switch fails; more cabling required than bus	Modern LAN (most common topology today — Ethernet with switch)
Ring	Devices connected in a closed loop; data travels in one direction (or both in dual ring). Each device acts as a repeater.	Equal access for all devices; predictable performance; no collisions	Break in ring = network failure (unless dual ring); adding devices interrupts the network; slower than star	Token Ring (legacy), SONET fiber optic backbone rings, some industrial networks
Mesh	Every device directly connected to every other device (full mesh) or to multiple others (partial mesh).	Highest fault tolerance — multiple paths; high bandwidth; failure of one link doesn't affect others	Very expensive (many cables/connections); complex to manage; $n(n-1)/2$ connections for n nodes	Internet backbone, military networks, mission-critical data center interconnects
Tree (Hierarchical)	Star networks connected together in a	Scalable — easy to expand; well-organized	Root node is a single point of failure; complex cabling	Large enterprise campus networks

Topology	Structure	Advantages	Disadvantages	Use Case
	hierarchy — branches extending from a central root.	hierarchy mirrors organizational structure		(building switches connected to floor switches connected to core switch)
Hybrid	Combination of two or more topologies.	Flexible — can optimize different parts of the network for their specific needs	Complex design and management	Real-world enterprise networks (always hybrid)

MESH CONNECTION COUNT — MATH

For a full mesh network with n nodes, the number of point-to-point links required is:

$$\text{Links} = n \times (n - 1) \div 2$$

For 5 nodes: $5 \times 4 \div 2 = \mathbf{10 \text{ links}}$

For 10 nodes: $10 \times 9 \div 2 = \mathbf{45 \text{ links}}$

For 20 nodes: $20 \times 19 \div 2 = \mathbf{190 \text{ links}}$

This exponential growth explains why full mesh is only practical for small networks or critical backbone segments — the cost of cabling and managing 190 connections for 20 nodes is prohibitive.

2. LAN, MAN, WAN

Type	Coverage	Typical Speed	Technology	Example
LAN (Local Area Network)	Single building or campus — up to ~1 km	100 Mbps to 10 Gbps	Ethernet (wired), Wi-Fi (wireless)	Office network, university campus network, factory floor IS network
MAN (Metropolitan Area Network)	City or metropolitan area — up to ~50 km	Up to 10 Gbps	Fiber optic, Ethernet MAN, WiMAX	City-wide government IS network, university networks spanning multiple campuses across a city, cable TV network in Dhaka
WAN (Wide Area Network)	Country, continent, or global	Varies widely	MPLS, leased lines, satellite,	Bank's nationwide ATM network connecting all branches; a multinational's IS connecting

Type	Coverage	Typical Speed	Technology	Example
		(kbps to Gbps)	4G/5G, Internet VPN	offices in Dhaka, London, and New York; the Internet itself (the global WAN)

3. Client-Server vs. Peer-to-Peer Architecture

Client-Server A powerful central server provides services (data, processing, storage) to multiple client devices. Clients request; server responds. Advantages: Centralized management, security, backup; scalable; consistent data. Disadvantages: Server = single point of failure; expensive server hardware; server must handle all load. Used by: all enterprise MIS (ERP, CRM, databases), web applications, email systems, MIS dashboards.	Peer-to-Peer (P2P) All nodes are equal — each can act as both client and server. Resources (files, processing) shared directly between peers without a central server. Advantages: No single point of failure; no server cost; highly scalable. Disadvantages: Difficult to secure and manage; inconsistent performance; not suitable for enterprise IS. Used by: file sharing (BitTorrent), blockchain networks, some VoIP systems (early Skype). Not suitable for enterprise MIS due to security and management challenges.
--	--

4. Circuit Switching vs. Packet Switching

Dimension	Circuit Switching	Packet Switching
Path	Dedicated physical circuit established for entire duration of communication	Data broken into packets; each packet independently routed through the network
Resource usage	Dedicated — bandwidth reserved even when no data is being transmitted (wasted capacity)	Dynamic — bandwidth shared; used only when packets are actually sent
Quality	Consistent, predictable — no delay variation once circuit established	Variable — packets may arrive out of order or be delayed (jitter)
Failure handling	Circuit failure = entire call dropped	Packets automatically rerouted around failures — highly resilient
Example	Traditional telephone call (PSTN) — a dedicated circuit is established from caller to receiver for the entire call duration	Internet data transmission — your web page request is split into packets, each routed independently, reassembled at destination

Dimension	Circuit Switching	Packet Switching
Use in MIS	Legacy voice networks; ISDN; some dedicated leased line MIS connections	All modern data networks including the Internet, LANs, WANs, VPNs — the dominant technology for MIS communication

5. VPN — Virtual Private Network

DEFINITION

A **VPN (Virtual Private Network)** creates an encrypted, authenticated tunnel through a public network (the Internet), allowing remote users or branch offices to access an organization's private MIS as if they were physically on the internal network.

- ▶ **Tunneling:** The VPN encapsulates private network packets inside encrypted outer packets for transmission over the public Internet. At the destination, the outer packet is removed and the inner packet delivered to the private network.
- ▶ **Encryption:** VPN uses protocols like IPSec (Internet Protocol Security) or SSL/TLS to encrypt all data in the tunnel — even if intercepted on the public Internet, the data is unreadable.
- ▶ **Authentication:** Users must authenticate (username + password, digital certificate, or multi-factor authentication) before VPN access is granted.
- ▶ **MIS relevance:** A Bangladeshi garment factory manager visiting a trade fair in Germany can securely access the factory's ERP system, check production status, and approve purchase orders through a VPN — as if sitting in the Dhaka head office.

6. SDN — Software Defined Networking

CONCEPT

Software Defined Networking (SDN) separates the network's control plane (decisions about where traffic is sent) from the data plane (actual forwarding of traffic), enabling network behaviour to be programmatically controlled from a central software controller rather than configured device-by-device.

- ▶ **Traditional networking:** Each router and switch independently makes routing decisions based on configuration. Changing network behaviour requires logging into each device individually.
- ▶ **SDN:** A central SDN controller has a global view of the network and programs all devices simultaneously via an API. Network changes (e.g., routing all MIS traffic through a low-latency path) can be made

programmatically in seconds.

- **MIS relevance:** SDN allows IS teams to dynamically prioritize MIS traffic (e.g., ensure the ERP system gets guaranteed bandwidth even during peak Internet usage), segment the network for security, and automate responses to failures — all through software without physical reconfiguration.

Telecom Network Types, Carriers & Business Value

CO2

Bloom's: Analyze

Teaching: Lecture + Quiz + Discussion

1. Types of Telecom Networks

Network Type	Description	Speed	Status
PSTN (Public Switched Telephone Network)	Traditional circuit-switched telephone network — the "plain old telephone system" (POTS). Uses copper wires and analog/digital switching.	Up to 56 kbps (dial-up data)	Legacy; being phased out as VoIP replaces it
ISDN (Integrated Services Digital Network)	Digital upgrade to PSTN — carries voice, data, and video over standard phone lines simultaneously. Two types: BRI (2 × 64 kbps channels) and PRI (30 × 64 kbps channels in Europe).	128 kbps (BRI) to 2 Mbps (PRI)	Legacy; largely replaced by broadband
DSL (Digital Subscriber Line)	Provides broadband Internet over existing copper telephone lines using frequencies above the voice range (so Internet and phone work simultaneously). ADSL: asymmetric (faster download than upload). VDSL: faster, shorter distance.	ADSL: 1–24 Mbps download; VDSL: up to 100 Mbps	Widely deployed; being replaced by fiber
Broadband Cable	Internet over the cable TV network's coaxial cables. Uses DOCSIS standard.	Up to 1 Gbps (DOCSIS 3.1)	Active; common in urban areas
Fiber Optic Broadband	FTTH (Fiber To The Home) / FTTB (Fiber To The Building) — optical fiber to the premises. Highest speeds available.	100 Mbps to 10 Gbps	Growing rapidly; future-proof
4G LTE	4th generation cellular — mobile broadband. LTE (Long Term Evolution) standard.	50–150 Mbps typical	Dominant mobile standard globally; mature
5G	5th generation cellular — see next section. Transformational for MIS and IoT.	Up to 20 Gbps (theoretical)	Deploying globally; Bangladesh deployment ongoing

2. 5G — Characteristics and MIS Implications

5G is not merely faster 4G — it is a fundamentally new network architecture designed to serve three distinct use cases (IMT-2020 specification):

eMBB
Enhanced Mobile Broadband

Extremely high data rates for humans consuming rich media — video streaming, VR, AR, high-resolution downloads. Target: 20 Gbps peak, 100 Mbps user experience.

MIS relevance: Real-time video-based quality inspection in factories; 4K video dashboards for remote monitoring; VR-based training systems.

URLLC
Ultra-Reliable Low Latency Communications

Latency below 1ms with 99.9999% reliability. For applications where delay or failure is catastrophic.

MIS relevance: Autonomous vehicle control, remote surgery, real-time industrial robot control, financial trading IS where microseconds matter.

mMTC
Massive Machine-Type Communications

Connecting up to 1 million devices per km² with low power consumption — the IoT enabler.

MIS relevance: Smart factory with thousands of sensors feeding production IS; smart city infrastructure monitoring; agricultural IoT reporting to farm management IS.

3. Telecom Carriers

Carrier Type	Description	Example
Common Carriers	Licensed telecom companies providing communication services to the public under government regulation. Must serve all customers at published tariffs without discrimination.	Bangladesh Telecommunications Company Ltd (BTCL), Grameenphone, Robi, Banglalink in Bangladesh; AT&T, Verizon in USA
Private Networks	Telecom infrastructure owned and operated by an organization exclusively for its own use. Provides highest security, performance, and control but at high cost.	Bangladesh Bank's own secure WAN connecting all branches; military communication networks; large industrial control networks
VAN Providers (Value-Added Network)	Private networks that provide specialized services beyond basic data transport — EDI translation, message formatting, security, guaranteed delivery, audit trails.	EDI service providers for RMG supply chains; healthcare data exchange networks (HL7); financial clearing networks (SWIFT for international banking)

4. Internet Service Providers (ISPs) — Tier Structure

○ Tier 1 ISPs — Internet Backbone

Global networks with direct connections to every other Tier 1 ISP through "peering" (free mutual traffic exchange). They own and operate the submarine cables, transcontinental fiber links, and major Internet exchange points. They do not pay anyone for Internet transit. Examples: AT&T, NTT, Tata Communications, PCCW. Bangladesh connects to Tier 1 ISPs via submarine cables (SEA-ME-WE 4, SEA-ME-WE 5) landing at Cox's Bazar.

○ Tier 2 ISPs — Regional Networks

Regional ISPs that peer with some networks for free but pay Tier 1 ISPs for transit to reach global destinations. Cover countries or major regions. Examples: Bangladesh Submarine Cable Company Limited (BSCCL) as the national bandwidth wholesaler; regional ISPs in Asia and Europe.

○ Tier 3 ISPs — Last-Mile Providers

Local ISPs that connect end users (homes, businesses, schools) to the Internet. Purchase transit from Tier 1 or Tier 2 ISPs. Examples: individual broadband ISPs in Bangladesh like Link3, BDCOM, Amber IT — they buy bandwidth from BSCCL and sell to end customers.

5. Business Value of Telecommunications

- ▶ **Cost reduction:** Telecom replaces physical movement — video conferencing replaces business travel; email replaces postal mail; electronic data exchange replaces paper-based document exchange. A single transatlantic flight costs \$1,50,000+; a video call costs near zero.
- ▶ **Global reach:** Organizations can access global markets, talent, and suppliers without physical presence. A Bangladeshi software startup can serve US clients entirely through telecom-enabled IS — no physical office required in the USA.
- ▶ **Real-time collaboration:** Teams across time zones collaborate on shared documents, IS, and projects in real-time. A Dhaka-based analyst and a London-based manager can review the same MIS dashboard simultaneously.
- ▶ **Supply chain integration:** Telecom enables real-time data exchange with suppliers, customers, and logistics partners — the foundation of modern supply chain IS and just-in-time operations.
- ▶ **New business models:** Telecom enables entirely new revenue models: telemedicine, remote education (bKash digital payments enabled by telecom), ride-sharing (Pathao, Shohoz — entirely telecom-dependent).

6. Cloud Networking — SD-WAN

SD-WAN — SOFTWARE DEFINED WAN

SD-WAN applies SDN principles to wide area networking, allowing organizations to use multiple cheaper Internet connections (broadband, 4G/5G) instead of expensive dedicated MPLS links to connect branch offices, while centrally managing traffic quality and security.

Dimension	Traditional MPLS WAN	SD-WAN
Cost	Very high — dedicated, privately managed circuits	Much lower — uses commodity Internet connections
Flexibility	Low — changes require carrier involvement, weeks to provision	High — software-defined changes in minutes
Performance	Guaranteed — SLA-backed quality	Managed — SD-WAN intelligently routes traffic over best available path in real-time
MIS suitability	Historically preferred for critical MIS due to guaranteed quality	Increasingly replacing MPLS as SD-WAN quality management matures; used by many large enterprises for MIS connectivity

1. Three Levels of Management — Information Needs

Managers at different levels require fundamentally different types of information. MIS must be designed to deliver the right information to the right level at the right time.

Attribute	Strategic (Top)	Tactical (Middle)	Operational (Lower)
Who	CEO, CFO, Board, VP level	Department heads, regional managers, functional managers	Supervisors, team leaders, front-line managers
Time horizon	Long-term: 3–10 years	Medium-term: months to 1 year	Short-term: daily to weekly
Aggregation level	Highly aggregated — company-wide KPIs, market trends	Moderately summarized — departmental performance vs. targets	Detailed — individual transactions, unit-level data
Internal/External	Primarily external — market data, competitor intelligence, economic indicators	Mix of internal and external — industry benchmarks + internal performance	Primarily internal — transaction records, production data, attendance
Accuracy required	Approximate — direction and trends matter more than precise figures	Reasonably accurate — departmental decisions require reliable data	Highly accurate — operational errors cause direct financial loss
Frequency	Periodic / event-triggered — monthly, quarterly, or when a strategic event occurs	Regular — weekly and monthly cycles	Continuous / real-time — every transaction must be processed immediately
IS used	EIS/ESS, DSS	MIS, DSS	TPS, Operational MIS

2. Anthony's Triangle

ROBERT N. ANTHONY'S FRAMEWORK

Anthony's Triangle (1965) maps management activities to three levels in a pyramid, each with distinct planning and control characteristics. It remains the foundational framework for understanding MIS design across organizational levels.

Level	Anthony's Activity	Description	IS Support
Top	Strategic Planning	Long-term goal setting, major resource allocation, defining organizational direction. Highly unstructured — relies heavily on external data and judgment.	EIS: aggregate trends, scenario analysis, competitor data, macroeconomic indicators
Middle	Management Control	Acquiring resources and using them efficiently to achieve strategic goals. Monitoring department performance, identifying variances, taking corrective action.	MIS: budget variance reports, departmental KPI dashboards, performance vs. target reports
Lower	Operational Control	Ensuring specific tasks are carried out effectively and efficiently. Day-to-day supervision, scheduling, quality control, transaction processing.	TPS: real-time transaction processing; operational IS for scheduling, inventory, attendance

3. Mintzberg's Managerial Roles

Henry Mintzberg identified 10 managerial roles grouped into three categories, each with specific information requirements that MIS must support:

INFORMATIONAL ROLES

- ▶ **Monitor:** Continuously scans the environment for information relevant to the organization. IS support: EIS dashboards, news feeds, market data alerts, social media monitoring IS.
- ▶ **Disseminator:** Transmits information received from outside or from subordinates to members of the organization. IS support: email systems, intranet, management reports, briefing tools.
- ▶ **Spokesperson:** Transmits information to outsiders on organization plans, policies, and results. IS support: reporting IS for investor relations, corporate communications IS, regulatory compliance reporting tools.

DECISIONAL ROLES

- ▶ **Entrepreneur:** Initiates and designs change in the organization — new projects, identifying opportunities. IS support: DSS for opportunity analysis, idea management IS, project IS.
- ▶ **Disturbance Handler:** Takes corrective action when organization faces unexpected crises. IS support: real-time exception alerts, operational IS status dashboards, crisis communication systems.
- ▶ **Resource Allocator:** Decides who gets what resources (budget, staff, equipment). IS support: resource management IS, budget allocation models, scheduling IS.

- ▶ **Negotiator:** Represents the organization in major negotiations with suppliers, customers, or partners. IS support: contract management IS, costing models, supplier performance data from IS.

INTERPERSONAL ROLES

- ▶ **Figurehead:** Performs ceremonial and symbolic duties. Limited IS support needed — but communications IS (scheduling, correspondence management) helps.
- ▶ **Leader:** Motivates and activates subordinates. IS support: performance management IS (tracking individual and team KPIs), HR-MIS for training and development tracking.
- ▶ **Liaison:** Maintains network of contacts outside the organization. IS support: CRM systems, contact management IS, LinkedIn integration.

4. Information Richness vs. Volume Trade-off

As information moves up the management hierarchy, its **volume** decreases but its **richness** (depth, context, analytical content) increases:

EXAMPLE — SAME DATA, THREE LEVELS

Operational Level (TPS): "Transaction #48821 — Customer: Rahman Textile | Product: SKU-4421 | Qty: 200 | Unit Price: ₹85 | Total: ₹17,000 | Date: 2025-06-10 | Payment: 30-day credit"

Tactical Level (MIS): "June 2025 Sales Report — Textile customers: 412 transactions, ₹3.2 crore total, 8% above June 2024, payment collection rate: 94%. Top 3 products: SKU-4421 (23%), SKU-3810 (18%), SKU-5902 (12%)."

Strategic Level (EIS): "Textile sector revenue growing at 8% YoY against market growth of 5% — we are gaining market share. However, credit exposure is ₹4.8 crore against a ₹4 crore policy limit. Recommend reviewing credit policy and exploring premium segment expansion."

5. Real-Time MIS — Dashboards, Live KPIs, Mobile Applications

- ▶ **Dashboards:** Visual displays of the most critical KPIs, updated in real-time or near-real-time. Replace traditional paper reports with always-current information accessible from any device. Example: a factory manager's dashboard showing current production rate, defect rate, machine downtime, and shift output — updated every 30 seconds.
- ▶ **Live KPI monitoring:** MIS continuously computes KPIs from streaming transaction data. Thresholds trigger automatic alerts when KPIs breach acceptable ranges — enabling immediate corrective action rather than waiting for the next scheduled report.

- **Mobile management applications:** MIS accessed via smartphone apps — extending decision support to wherever the manager is. A regional sales manager can review branch performance, approve discounts, and respond to exception alerts from the road. Bangladesh example: Grameenphone management using mobile BI apps to monitor real-time network performance and subscriber metrics.

1. Decision Making Process — Expanded Simon Model

The expanded Simon model adds two phases to the original four, creating a more complete cycle:

Phase	Activity	Information Needed	IS Support
Intelligence	Problem/opportunity identification; scanning the environment	Exception reports, trend data, anomaly alerts, external market data	MIS exception reports, EIS dashboards, environmental scanning IS
Design	Developing and analyzing alternative solutions	Historical data for modelling, cost data, capacity data, market projections	DSS models, simulation tools, financial planning IS
Choice	Selecting the best alternative using criteria and models	Comparison of alternatives on key metrics, risk analysis, sensitivity analysis	DSS optimization models, decision matrices, risk analysis tools
Review	Checking the chosen alternative before full commitment	Feasibility data, pilot results, stakeholder feedback	Prototype systems, pilot IS, simulation results
Implementation	Executing the decision and monitoring outcomes	Real-time operational data, KPI tracking, budget actuals vs. plan	TPS for execution, MIS for monitoring, dashboard alerts for variances

2. Decision Environments — Certainty, Risk, Uncertainty

Certainty

The outcome of each alternative is known in advance with complete confidence. The decision is about selecting the alternative with the best known outcome.

Example: Choosing between two suppliers whose prices, quality records, and delivery times are fully

Risk

Multiple possible outcomes exist for each alternative, and the probability of each outcome can be estimated from historical data or expert judgment.

Example: Launching a new product — historical data suggests 60% chance of success (profit ₹50 lakh), 40% chance of failure (loss ₹20 lakh). DSS models can compute expected values under risk.

known from contracts. MIS can automate decisions under certainty — no human judgment needed.

Uncertainty

Multiple possible outcomes exist, but probabilities cannot be estimated — the decision maker does not know enough to assign reliable probabilities. The most challenging environment for MIS support.

Example: Deciding whether to enter a new country market with no historical data; deciding how to respond to an unprecedented global event (e.g., a pandemic).
EIS provides context; human judgment dominates.

3. Expected Value (EV) — Decision Under Risk

$$EV = \sum (P_i \times V_i)$$

where: P_i = probability of outcome i / V_i = value (payoff) of outcome i /
 Σ = sum over all outcomes

EXPECTED VALUE — WORKED EXAMPLE 1

Scenario: A company must decide whether to launch Product A or Product B. Historical market research gives the following probability distributions:

Product A:

- High demand ($P = 0.50$): Profit = ₹60,00,000
- Medium demand ($P = 0.30$): Profit = ₹25,00,000
- Low demand ($P = 0.20$): Loss = -₹10,00,000

$$EV(A) = (0.50 \times 60,00,000) + (0.30 \times 25,00,000) + (0.20 \times -10,00,000)$$
$$= 30,00,000 + 7,50,000 - 2,00,000 = ₹35,50,000$$

Product B:

- High demand ($P = 0.30$): Profit = ₹80,00,000
- Medium demand ($P = 0.40$): Profit = ₹30,00,000
- Low demand ($P = 0.30$): Loss = -₹20,00,000

$$EV(B) = (0.30 \times 80,00,000) + (0.40 \times 30,00,000) + (0.30 \times -20,00,000)$$
$$= 24,00,000 + 12,00,000 - 6,00,000 = ₹30,00,000$$

Decision: Choose Product A ($EV = ₹35.5 \text{ lakh} > EV \text{ of B} = ₹30 \text{ lakh}$)

Note: EV maximization is a rational decision rule, but it ignores risk aversion. A company that cannot afford the worst-case loss of Product B (₹20 lakh) might prefer Product A even if B has higher upside — risk tolerance must be considered alongside EV.

4. Decision Trees

A **decision tree** is a graphical tool for structuring multi-stage decisions under risk. It shows alternatives, chance events, probabilities, and payoffs in a tree structure that is solved by "rolling back" from right to left (from outcomes to decisions).

DECISION TREE SYMBOLS

- ▶ **Square node (□):** Decision node — the decision maker chooses which branch to follow.
- ▶ **Circle node (○):** Chance node — nature determines which outcome occurs, with given probabilities.
- ▶ **Branches:** Lines from decision nodes represent alternatives; lines from chance nodes represent possible outcomes.
- ▶ **Leaf values:** The payoff (profit/loss) at the end of each path through the tree.

DECISION TREE — WORKED EXAMPLE

Scenario: A software company must decide: (1) Build a new IS in-house, or (2) Buy a commercial package. If built in-house, it might succeed ($P=0.7$, value +₹40L) or fail ($P=0.3$, value -₹15L). If bought, the package fits well ($P=0.6$, value +₹25L) or fits poorly ($P=0.4$, value -₹5L).

□ DECISION — BUILD IN-HOUSE | ○ Chance — Success ($P=0.70$) → +₹40,00,000
| — Failure ($P=0.30$) → -₹15,00,000 | | EV = $(0.70 \times 40) + (0.30 \times -15) = 28 - 4.5 = ₹23.5 \text{ lakh}$ | — BUY PACKAGE ○ Chance — Good fit ($P=0.60$) → +₹25,00,000
| — Poor fit ($P=0.40$) → -₹5,00,000 EV = $(0.60 \times 25) + (0.40 \times -5) = 15 - 2 = ₹13 \text{ lakh}$ ▶ DECISION: Build in-house (EV ₹23.5L > ₹13L)

Roll-back method: compute EV at each chance node, then select the decision branch with the highest EV.

5. Bounded Rationality — Herbert Simon

CONCEPT

Bounded rationality is Herbert Simon's theory that real-world decision makers cannot achieve the perfectly rational optimization assumed by classical economic models — because human cognitive capacity, time, and available information are all limited (bounded).

- ▶ **Satisficing vs. Optimizing:** A truly rational decision maker would evaluate all possible alternatives and select the absolute best one (optimizing). A bounded rationality decision maker evaluates alternatives sequentially and stops when they find one that is "good enough" — meeting a threshold of acceptability. This is satisficing (satisfying + sufficing).
- ▶ **MIS implication:** MIS cannot make humans perfectly rational, but it can expand the bounds — providing more information, faster analysis, and better decision models. DSS is specifically designed to extend bounded rationality by processing more data and alternatives than a human could manually evaluate.
- ▶ **Example:** A procurement manager satisfices by selecting the first supplier whose price, quality, and delivery time all meet minimum acceptable thresholds — rather than evaluating every supplier in the market. A DSS can evaluate thousands of suppliers and flag the truly optimal choice, extending the manager's bounded rationality.

6. Cognitive Biases in Decision Making

Bias	Description	IS / MIS Counter-measure
Anchoring Bias	Over-relying on the first piece of information encountered. A manager who sees last year's ₹100 lakh budget first will anchor all subsequent budget discussions around that figure, even if circumstances have changed dramatically.	DSS presents multiple reference points and benchmarks simultaneously, preventing fixation on a single anchor. Historical ranges and industry comparisons help contextualize figures.
Confirmation Bias	Seeking and favouring information that confirms existing beliefs while ignoring contradictory evidence. A manager convinced a strategy is working will dismiss MIS reports showing otherwise.	MIS exception reports and anomaly detection force managers to confront disconfirming data. Dashboards present all KPIs objectively regardless of whether they support current beliefs.
Availability Heuristic	Judging probability based on how easily examples come to mind. A manager who recently witnessed a supplier failure will overestimate the probability of future failures from all suppliers.	DSS provides actual historical frequency data rather than relying on what's mentally salient. Statistical base rates from the IS database override mental availability.
Overconfidence	Decision makers are systematically overconfident in their predictions.	MIS can present historical accuracy of past forecasts — showing managers their track record of over/underestimation.

Bias	Description	IS / MIS Counter-measure
	Managers routinely underestimate project costs and overestimate demand.	Probabilistic DSS models quantify uncertainty rather than presenting false precision.

1. DSS — Definition

DEFINITION

A **Decision Support System (DSS)** is an interactive, computer-based IS that helps decision makers use data and models to identify and solve semi-structured and unstructured problems and make decisions. DSS supports — but does not automate or replace — human judgment.

Key characteristics that distinguish DSS from MIS:

- ▶ **Interactive:** The manager actively engages with the DSS — running what-if scenarios, changing parameters, exploring data — rather than passively receiving a scheduled report.
- ▶ **Model-driven:** DSS includes analytical and mathematical models (statistical, financial, optimization) that transform data into decision-relevant insights.
- ▶ **Supports semi-structured decisions:** MIS handles structured decisions (routine reporting); DSS handles decisions with both structured and unstructured components.
- ▶ **Flexible:** Managers can customize queries, models, and analyses without involving the IT department.

2. Types of DSS

Model-Driven DSS

Primarily uses mathematical models (financial, statistical, optimization, simulation) to analyze data and generate decision recommendations. The model is the core component.

Examples: Financial planning DSS (spreadsheet-based what-if models), production scheduling optimization DSS, portfolio optimization DSS, supply chain simulation DSS.

Data-Driven DSS

Emphasizes access to and analysis of large data sets — a data warehouse or database with powerful query and analysis tools. Includes OLAP (Online Analytical Processing) and data mining.

Examples: A retail chain's sales analysis DSS that allows managers to slice and dice sales data by product, region, time period, and customer segment in any combination.

Communications-Driven DSS (GDSS)

Facilitates collaborative decision making among groups of people — the Group DSS (GDSS) covered in

Knowledge-Driven DSS

Incorporates expert knowledge (rules, heuristics) to recommend decisions in a specific domain.

Class 24. Network-based tools for brainstorming, voting, and consensus building.

Examples: Electronic meeting systems, collaborative planning tools, virtual war rooms for strategic planning sessions.

Resembles Expert Systems — "if condition X then recommend action Y."

Examples: Medical diagnosis support systems (recommend treatments based on symptoms + test results), loan approval DSS (assess creditworthiness based on expert rules), fraud detection systems.

3. DSS Components

Data Management Component

Stores and provides access to the data the DSS needs — internal transaction data, historical records, external market data. Includes the DSS database and a database management system (DBMS).

May be a dedicated DSS database separate from the operational TPS database to avoid performance interference, or a data warehouse that consolidates data from multiple sources.

Model Management Component

Houses the mathematical and analytical models (the "model base"): financial models, statistical models, optimization algorithms, simulation models, forecasting models. A model management system (MMS) allows models to be stored, retrieved, and combined.

User Interface (Dialogue Manager)

Enables the manager to interact with the DSS — entering parameters, selecting models, requesting analyses, and viewing results in meaningful formats (charts, tables, dashboards). Must be usable by non-technical managers without IT assistance.

4. Analytical Models in DSS

STATISTICAL MODELS

Applied to large datasets to identify patterns, correlations, and trends. Examples: regression analysis (identifying which factors drive sales), time-series analysis (forecasting future demand based on historical trends), clustering (grouping customers by behaviour patterns for targeted marketing).

FINANCIAL MODELS

Mathematical representations of financial relationships — P&L statements, cash flow models, break-even analysis, DCF (Discounted Cash Flow) models. Enable what-if analysis on financial parameters.

OPTIMIZATION MODELS

Find the best solution from a set of feasible alternatives — maximizing profit, minimizing cost, or minimizing time subject to constraints. Linear programming (LP) is the most common technique.

FORECASTING MODELS

Project future values based on historical data. Examples: moving average, exponential smoothing, ARIMA. Essential for demand forecasting, budget planning, and capacity planning IS.

5. What-If Analysis

DEFINITION

What-if analysis allows a decision maker to change the value of one or more input variables in a model and observe how the output changes — exploring "what would happen if..." scenarios without real-world consequences.

WHAT-IF ANALYSIS — SENSITIVITY TABLE (BREAK-EVEN)

Scenario: A company sells a product. Fixed costs: ₪5,00,000/month. Variable cost: ₪120/unit. Selling price: ₪200/unit. Contribution margin = ₪200 – ₪120 = ₪80/unit.

Break-even quantity = Fixed Costs ÷ Contribution Margin = 5,00,000 ÷ 80 = 6,250 units/month

The DSS what-if table shows how break-even changes as selling price or variable cost changes:

Selling Price (₪)	Variable Cost (₪)	Contribution Margin (₪)	Break-Even Units	Profit at 8,000 units
180	120	60	8,333	-₪33,333 (loss)
200	120	80	6,250	+₪1,40,000
220	120	100	5,000	+₪3,00,000
200	110	90	5,556	+₪2,20,000
200	130	70	7,143	+₪60,000

The DSS instantly computes all scenarios — a manager can see that a ₪20 price increase changes break-even from 6,250 to 5,000 units and triples the profit at 8,000 units. This level of scenario exploration would take hours manually; DSS delivers it in seconds.

6. Goal-Seeking Analysis

DEFINITION

Goal-seeking analysis is the reverse of what-if analysis. Instead of "if I change this input, what happens to the output?", goal-seeking asks "what input value do I need to achieve a desired output?"

GOAL-SEEKING — WORKED EXAMPLE

Scenario: The finance director wants to achieve a profit of ₺5,00,000 next month. Fixed costs are ₺5,00,000; variable cost per unit is ₺120.

Using the same contribution margin structure:

Required Revenue = Fixed Costs + Target Profit = 5,00,000 + 5,00,000 = ₺10,00,000

If selling price = ₺200: Required units = 10,00,000 ÷ 200 = **5,000 units**

The DSS (goal-seeking function) can also ask: "At what selling price do we achieve ₺5,00,000 profit if we can only sell 7,000 units?"

Revenue needed = 10,00,000; Quantity = 7,000; Required price = 10,00,000 ÷ 7,000 = **₺142.86/unit**
minimum (plus variable cost recovery: ₺120 + ₺71.43 = ~₺191.43/unit minimum selling price)

7. DSS Applications

Domain	DSS Application	Models Used
Financial Planning	Budget allocation, investment appraisal (NPV, IRR), cash flow forecasting, capital structure optimization	DCF models, scenario analysis, what-if sensitivity tables
Production Scheduling	Optimizing machine scheduling to minimize idle time and maximize throughput; workforce scheduling to minimize overtime cost	Linear programming, simulation, constraint satisfaction
Portfolio Management	Selecting the optimal mix of investments (stocks, bonds, projects) to maximize return for a given risk level	Markowitz mean-variance optimization, Monte Carlo simulation, risk-return models
Supply Chain	Inventory optimization (reorder points, safety stock levels), supplier selection, logistics route optimization	EOQ model, demand forecasting, linear programming for routing
Marketing	Market segmentation, campaign ROI analysis, pricing optimization, customer lifetime value modelling	Regression, clustering, CLV models, A/B test analysis
Healthcare	Resource allocation (beds, staff, equipment), diagnosis support, treatment protocol	Knowledge-driven rules, statistical models, queueing theory

CASE STUDY**BRAC Bank Bangladesh — DSS for Credit Risk Assessment**

Background: BRAC Bank is one of Bangladesh's largest SME-focused banks, with a large loan portfolio to small and medium enterprises. Credit risk assessment — deciding whether to approve a loan and at what terms — is a classic semi-structured decision requiring both data analysis (financial ratios) and human judgment (business viability, character of the borrower).

Pre-DSS situation: Loan officers manually reviewed applications, calculated financial ratios on spreadsheets, and made subjective recommendations based on personal experience. Assessment quality varied widely between officers, and the process took 7–10 working days per application.

DSS components deployed:

- **Data management:** CIB (Credit Information Bureau) data integrated, applicant financial statements entered into DSS database, historical loan repayment records.
- **Model management:** Credit scoring model (logistic regression trained on 5 years of loan outcomes), cash flow adequacy model ($\text{DSC Ratio} = \text{Net Operating Income} \div \text{Total Debt Service}$), collateral valuation model.
- **User interface:** Loan officers enter application data; DSS displays a composite credit score, key financial ratios with traffic-light indicators, and a recommended credit limit.

DSS Output (example):

- Credit Score: 724/1000 (Good)
- Debt Service Coverage Ratio: 1.45 (acceptable; threshold is 1.25)
- Current Ratio: 1.8 (adequate liquidity)
- DSS Recommendation: Approve up to ~~1~~50 lakh at 11% p.a., 36-month term

Outcome: Assessment time reduced from 7–10 days to 2–3 days. Default rate on DSS-assessed loans 18% lower than manually assessed loans. Loan officer time freed for relationship management and complex cases requiring deeper judgment.

Lesson: DSS does not replace the loan officer — the officer reviews the DSS recommendation and makes the final decision. DSS extends bounded rationality, ensuring all relevant data is considered consistently and objectively.

